

First Phase of Ethical Hacking – Reconnaissance



What is Reconnaissance?

Reconnaissance consists of techniques that involve actively or passively gathering information that can be used to support targeting. Such information may include details of the victim organization, infrastructure, or staff/personnel.

Primary focus is on gathering valuable information about a target system or network.

The goal of this phase is to collect as much data as possible to understand the target's structure, vulnerabilities, and weaknesses, which can later be exploited in the subsequent stages of a security assessment



Reconnaissance vs Footprinting vs Information Gathering

- Recon - The broadest term, encompassing all activities to gather preliminary information about a target before an attack or penetration test. Includes Footprinting and Information Gathering.
- Footprinting - A structured and targeted phase of reconnaissance that focuses on creating a blueprint of the target system or organization. To gather specific information like domain names, IP address ranges, open ports, and system architecture.
- Information Gathering - To collect any useful data — not just technical but also organizational or personal (e.g., employee emails, software used, etc.).

Comparison Table

Aspect	Reconnaissance	Footprinting	Information Gathering
Scope	Broadest	Specific and targeted	General
Relation	Umbrella term	Subset of reconnaissance	Part of recon and beyond
Purpose	Understand target	Create network blueprint	Collect useful data
Types	Passive / Active	Passive / Active	OSINT, tech info, etc.
Example	Monitoring network traffic	Finding open ports via Nmap	Searching LinkedIn for staff

Types of Reconnaissance

Passive Reconnaissance

- No direct interaction with the target.
- Information gathered from public sources.
- Examples - WHOIS lookups, DNS queries, Social Media/OSINT (Open Source Intelligence)

Active reconnaissance

- Direct interaction with the target system.
- Involves tools and techniques like port scanning or network mapping.
- Examples - Ping sweeps, DNS zone transfers, Nmap scanning

Key Steps in the Reconnaissance Process

- Most organizations share a tremendous amount of information and data through various channels, including their websites and social media pages, their employees, and even their help desks.



What information to collect?

- Gather Victim Identity Information
 - Credentials
 - Email Addresses
 - Employee Names
- Gather Victim Org Information
 - Business Relationships
 - Determine Physical Locations
 - Identify Business Tempo
 - Identify Roles
- Gather Victim Network Information
 - DNS
 - Domain Properties
 - IP Addresses
 - Network Security Appliances
 - Network Topology
 - Network Trust Dependencies
- Search Open Technical Databases
 - DNS/Passive DNS
 - Whois
 - Digital Certificates
 - Social Media

What information to collect?

- Gather Victim Host Information
 - Client Configurations
 - Firmware
 - Hardware
 - Software
- Internet Hosted Services
 - Web Services
 - Email Services
 - FTP, VPN, DNS
 - Remote Login Services

Key methods of passive reconnaissance

- Search Engines (Google Dorking)
- WHOIS Lookups
- DNS Queries (Domain Name System Queries)
- Social Media and Open Source Intelligence (OSINT)
- Public Data Repositories
- Network and Web Mapping (Using Public Tools)

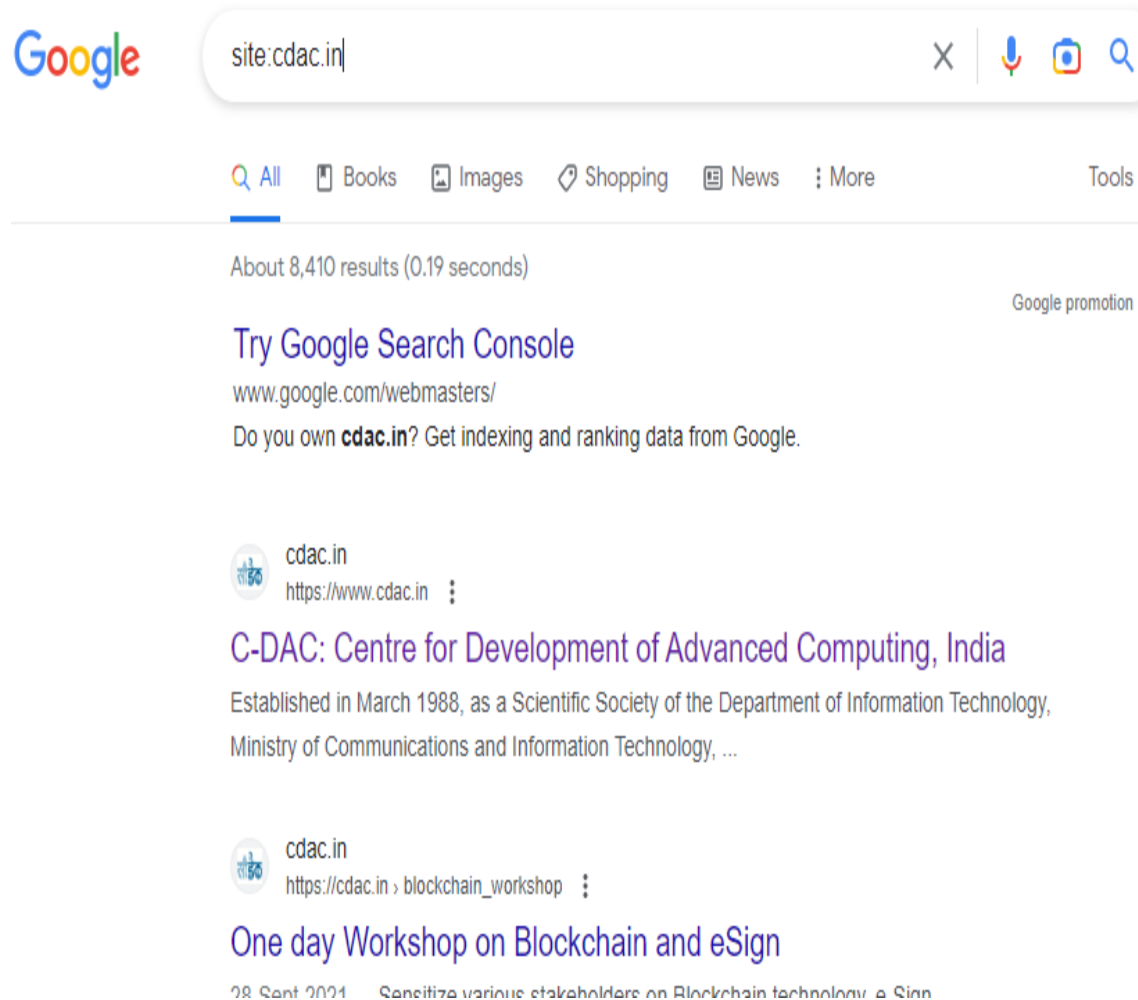
Passive/Active reconnaissance using Google Dork

- Google Dork uses advanced search queries to extract sensitive information from public web pages indexed by Google.
- leverages Google's advanced search operators to retrieve specific types of information that are often unintentionally exposed online, such as -
 - Sensitive files (e.g., .doc, .pdf, .xls)
 - Website configurations
 - Login pages
 - Vulnerable servers
 - Email addresses

“google dork” Search Parameters

Keyword	Function	Example
site -	provides a list of every URL from a website or domain that has been indexed.	site - ieee.org
filetype -	based on the specified file extension, returns several types of files	filetype - ppt
inurl -	looks in the URL for a certain phrase	Inurl - admin.php
allinurl -	returns results whose URL contains every character you provide. Searches for occurrences of all the keywords given.	allinurl - abcde
intext -	finds websites with certain characters or strings in their content.	intext - “CDAC Bangalore”
	Shows all sites that contain either or both specified words in the query	CDAC Bangalore
+	Concatenates words to detect pages using more than one specific key	CDAC+ Bangalore
before/after	Used to search within a particular date range.	filetype - pdf & (before - 2020-01-01 after - 2020-01-01)

site -



By using the "site - " operator along with a keyword, you can find information that is only present on a particular website.

site - cdac.in security

You can also use more advanced operators and search terms in combination with the "site - " operator to perform even more targeted searches.

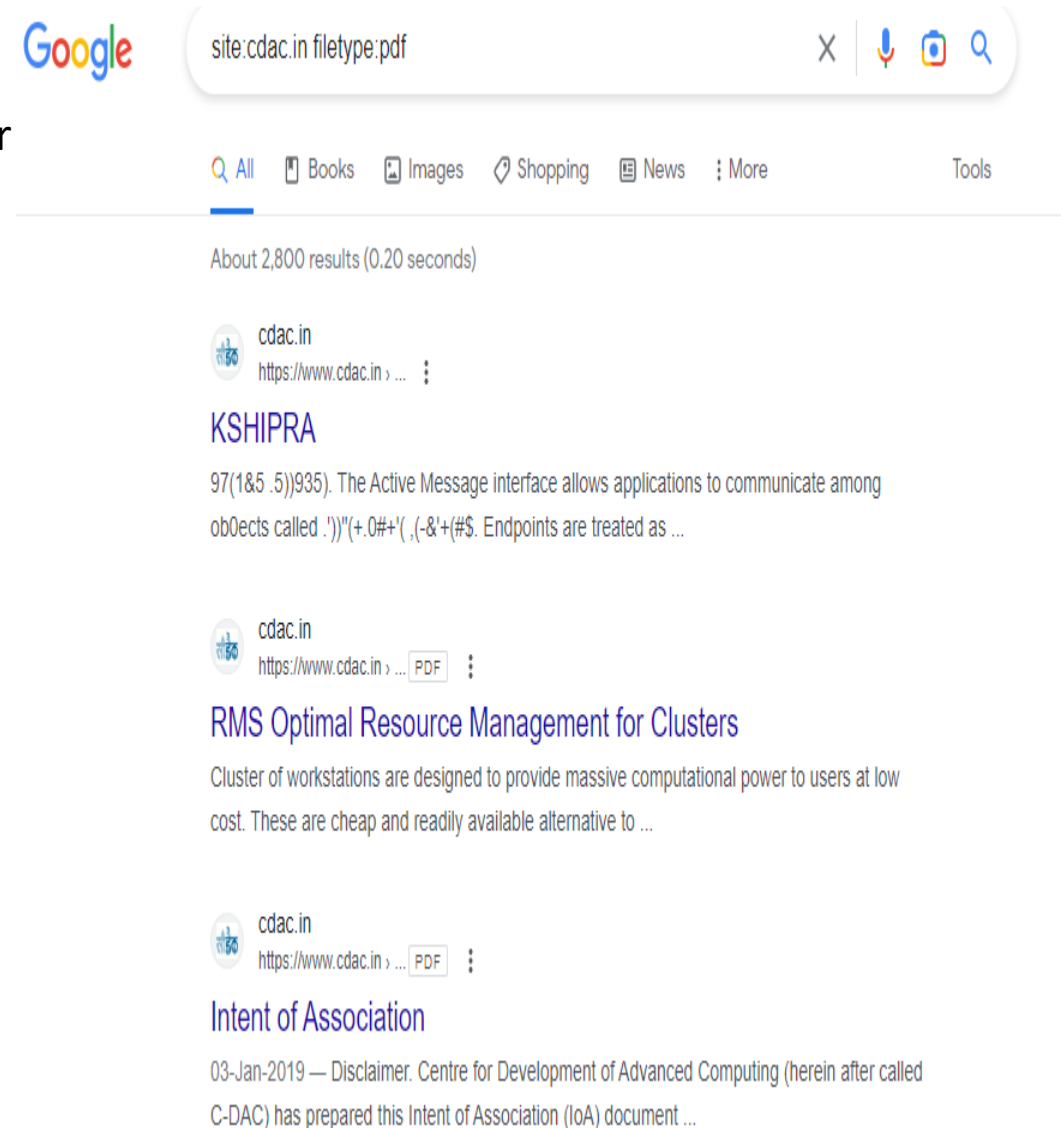
site - cdac.in intitle - login

Filetype -

It is a specific search query that uses the "filetype - " operator to search for files of a particular type or format on the internet.

It is useful to narrow down your search results to specific file types.

site - cdac.in filetype - pdf



The screenshot shows a Google search interface. The search bar contains the query "site:cdac.in filetype:pdf". Below the search bar, there are navigation links for "All", "Books", "Images", "Shopping", "News", and "More". The search results are displayed below the navigation links. The first result is titled "KSHIPRA" and is a PDF document. The second result is titled "RMS Optimal Resource Management for Clusters" and is also a PDF document. The third result is titled "Intent of Association" and is a PDF document. Each result includes a snippet of text from the document.

Google

site:cdac.in filetype:pdf

Search

All Books Images Shopping News More Tools

About 2,800 results (0.20 seconds)

cdac.in
https://www.cdac.in > ...

KSHIPRA

97(1&5 .5))935). The Active Message interface allows applications to communicate among objects called .')")(+.0#+'((-&'!+(\$\$. Endpoints are treated as ...

cdac.in
https://www.cdac.in > ... PDF

RMS Optimal Resource Management for Clusters

Cluster of workstations are designed to provide massive computational power to users at low cost. These are cheap and readily available alternative to ...

cdac.in
https://www.cdac.in > ... PDF

Intent of Association

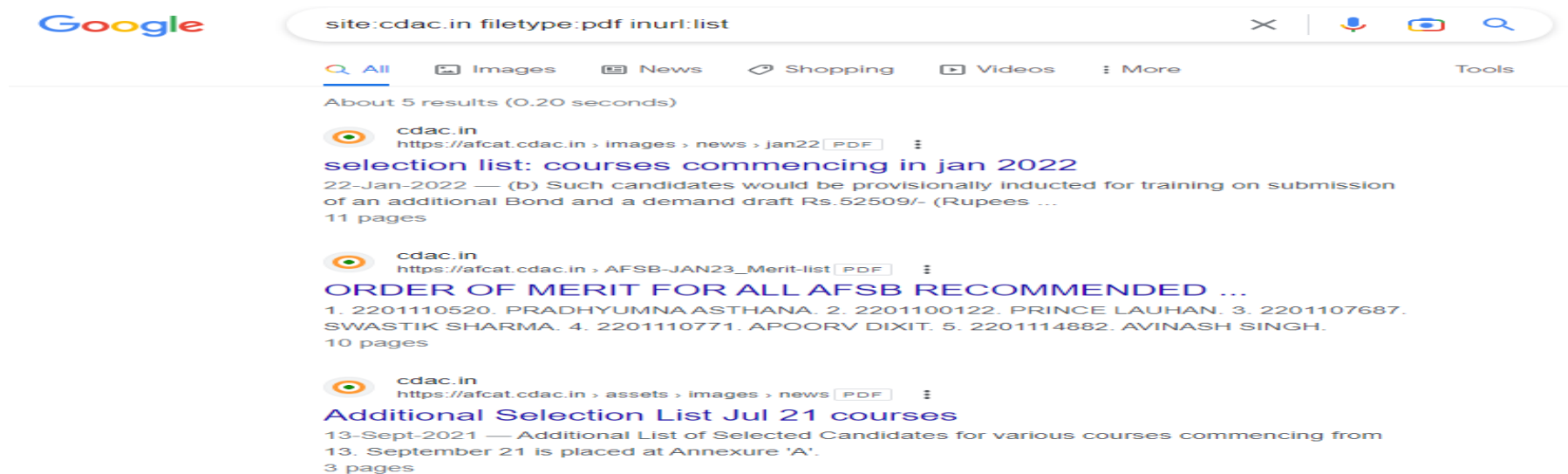
03-Jan-2019 — Disclaimer. Centre for Development of Advanced Computing (herein after called C-DAC) has prepared this Intent of Association (IoA) document ...

Inurl -

It is used to find web pages with a particular keyword or term in their URL

Useful for narrowing down search results to specific URLs or directories that contain the specified keyword

You can also use the "inurl" operator in combination with other search terms or operators to perform more targeted searches.



Intext -

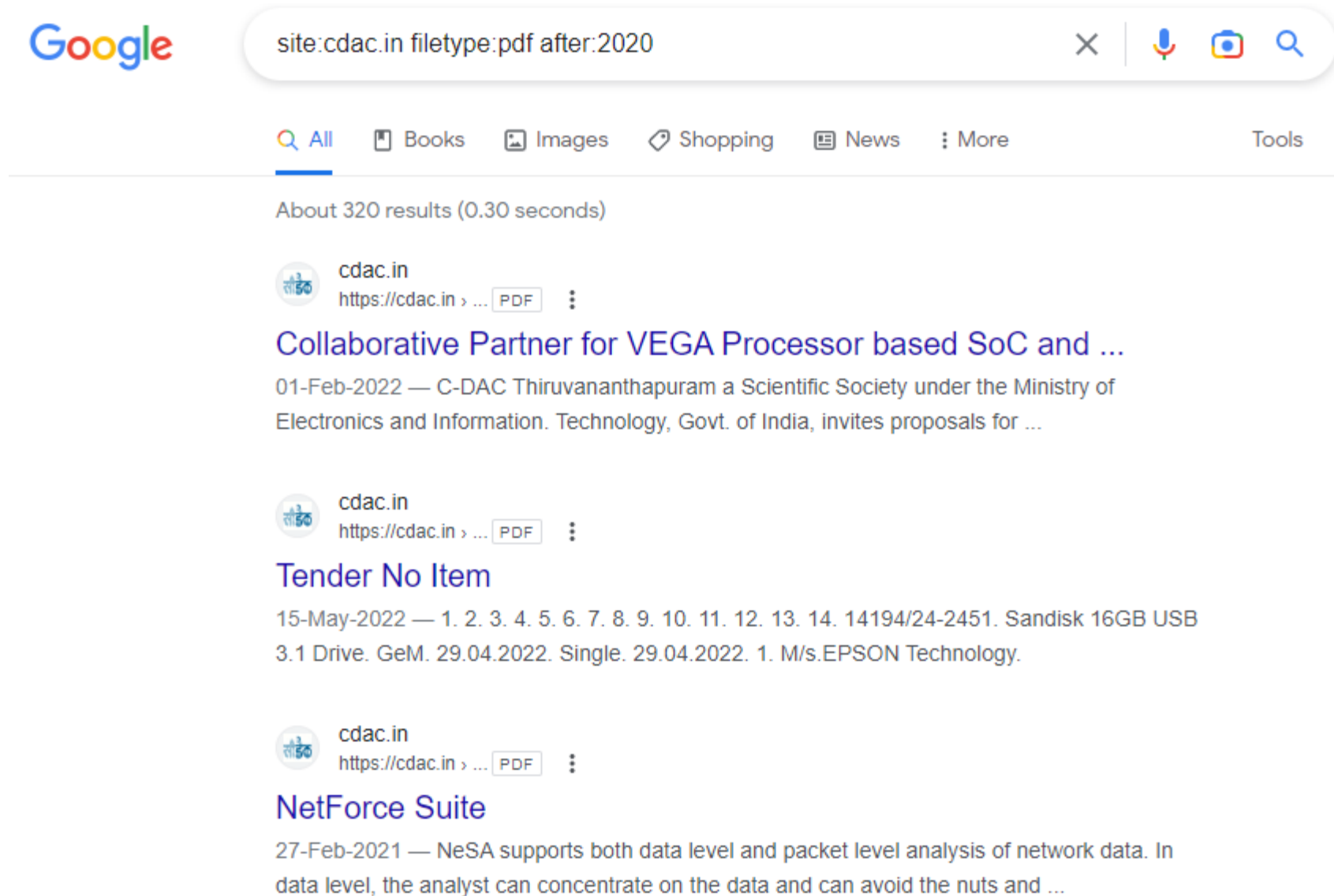
Used to find web pages containing a particular keyword or phrase in their content (body text).

This operator is useful for narrowing down search results to pages that include the specified keyword within their text.

intext - keyword



Before/after



The screenshot shows a Google search interface. The search bar contains the query 'site:cdac.in filetype:pdf after:2020'. Below the search bar, there are tabs for 'All', 'Books', 'Images', 'Shopping', 'News', and 'More', with 'All' being the selected tab. The search results are displayed below the tabs, showing 'About 320 results (0.30 seconds)'. The first three results are from 'cdac.in' and are PDF files. The first result is titled 'Collaborative Partner for VEGA Processor based SoC and ...' and is dated '01-Feb-2022'. The second result is titled 'Tender No Item' and is dated '15-May-2022'. The third result is titled 'NetForce Suite' and is dated '27-Feb-2021'.

Google

site:cdac.in filetype:pdf after:2020

× | 🔊 📷 🔍

🔍 All 📖 Books 🖼️ Images 🛒 Shopping 📰 News ⋮ More Tools

About 320 results (0.30 seconds)

 cdac.in
https://cdac.in > ... PDF ⋮

Collaborative Partner for VEGA Processor based SoC and ...

01-Feb-2022 — C-DAC Thiruvananthapuram a Scientific Society under the Ministry of Electronics and Information. Technology, Govt. of India, invites proposals for ...

 cdac.in
https://cdac.in > ... PDF ⋮

Tender No Item

15-May-2022 — 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 14194/24-2451. Sandisk 16GB USB 3.1 Drive. GeM. 29.04.2022. Single. 29.04.2022. 1. M/s.EPSON Technology.

 cdac.in
https://cdac.in > ... PDF ⋮

NetForce Suite

27-Feb-2021 — NeSA supports both data level and packet level analysis of network data. In data level, the analyst can concentrate on the data and can avoid the nuts and ...

Google Dorks Examples

1. To Find details of FTP servers

intitle - "index of" inurl - ftp

2. To locate all the HTTP websites

intitle - "index of" inurl - http - // after - 2020

3. To find the list of online cameras

Intitle - “webcamxp 5”

Exploit Database

EXPLOIT
DATABASE

Category

Files Containing Passwords

Clear

Author

Begin typing...

Google Hacking Database

Show 15

Quick Search

Date Added	Dork	Category	Author
2024-08-23	site:github.com "BEGIN OPENSSSH PRIVATE KEY"	Files Containing Passwords	kstrawn0
2024-08-23	ext:nix "BEGIN OPENSSSH PRIVATE KEY"	Files Containing Passwords	kstrawn0
2024-07-04	intitle:index of /etc/ssh	Files Containing Passwords	Shivam Dhingra
2024-07-04	intext:"aws_access_key_id" intext:"aws_secret_access_key" filetype:json filetype:yaml	Files Containing Passwords	Joel Indra
2022-06-15	intitle:"Index of" pwd.db	Files Containing Passwords	Muhammad Al-Amin
2022-06-15	intitle:"Index of" htpasswd	Files Containing Passwords	Muhammad Al-Amin
2021-11-15	site:controlc.com intext:"password"	Files Containing Passwords	Anirudh Kumar Kushwaha
2021-11-15	site:reentry.co intext:"password"	Files Containing Passwords	Anirudh Kumar Kushwaha

Gather Victim Identity Information – Using Google Hacks

Credentials (Usernames, Passwords)

inurl:"/ftp.txt" OR inurl:"/config.txt" OR inurl:"/passwd" OR inurl:"credentials.txt"
intitle:"index of" "password"
filetype:log intext:password
intext:"login" intext:"password" filetype:txt

Email Addresses

intext:"@example.com"
site:example.com "@example.com"
filetype:xls intext:"email"
filetype:txt intext:"email"
site:example.com intext:"email"

Employee Names

site:example.com filetype:pdf OR filetype:docx "employee list"
site:linkedin.com/in "company name" AND "employee"
intext:"meet the team" site:example.com
filetype:xls intext:"name" AND intext:"email"

Use site: to restrict to a specific domain.

Combine with filetype: to extract structured data (PDF, DOCX, XLS).

Use " " to force exact match phrases

Gather Victim Org Information

Business Relationships

site:example.com "partner" OR "client" OR "vendor" OR "affiliate"

site:example.com filetype:pdf "memorandum of understanding" OR "contract" OR "agreement"

site:sec.gov "example inc" (for public companies – financial reports)

"example.com" AND "powered by" OR "in partnership with" OR "supported by"

Determine Physical Locations

site:example.com "head office" OR "our location" OR "contact us"

site:example.com intext:"address" OR intext:"headquarters" filetype:pdf

"company name" + "Google Maps"

filetype:xls OR filetype:csv "office" "address" site:example.com

Identify Business Tempo (Work Schedule, Events, Activity Cycles)

site:example.com "business hours" OR "support hours" OR "maintenance schedule"

site:example.com intext:"scheduled downtime" OR "holiday closure"

"example.com" "calendar of events" OR "event schedule"

site:example.com "Q1 report" OR "quarterly update"

Identify Roles (Key Staff, Org Structure, Job Functions)

site:example.com "our team" OR "leadership team" OR "management"

site:linkedin.com/in ("company name" AND "CTO" OR "engineer" OR "sales")

filetype:pdf OR filetype:doc "organizational chart" OR "org structure"

site:example.com careers OR "job openings" OR "we are hiring"

Gather Victim Network Information

DNS Information

Google Dorks won't give DNS records directly, but they can uncover DNS-related files or misconfigurations.

filetype:txt site:example.com intext:"DNS"

filetype:conf OR filetype:txt inurl:dns OR inurl:named.conf

intitle:"index of" "zone file" site:example.com

site:pastebin.com intext:"DNS" "example.com"

Network Security Appliances - Find firewalls, VPNs, IDS/IPS, etc., via login portals or leaked docs

intitle:"SonicWALL" OR intitle:"Fortinet" OR intitle:"Cisco ASA"

site:example.com filetype:pdf "firewall configuration"

inurl:/admin OR inurl:/vpn intitle:"login" site:example.com

intitle:"login" AND "Palo Alto Networks"

Network Topology

filetype:vsd OR filetype:pdf "network topology" site:example.com

filetype:drawio OR filetype:gliffy OR filetype:xml "network diagram"

"internal network layout" OR "LAN diagram" site:example.com

intitle:"index of" "network diagram"

Shodan - The search engine for Hackers

One of the most essential tools for many cybersecurity professionals, including penetration testers, bug hunters, and security researchers, is Shodan. With the help of this application, it is possible to find servers, cameras, control systems, and databases that are accessible over the internet.

The screenshot shows the Shodan website dashboard. At the top, there is a navigation bar with links for Shodan, Maps, Images, Monitor, Developer, and More... Below this is a dark header with the Shodan logo, navigation links (Explore, Downloads, Pricing), a search bar, and an Account link. The main content area features a 'Dashboard' section with three cards: 'Getting Started' (with links to 'What is Shodan?', 'Search Query Fundamentals', and 'Working with Shodan Data Files'), '>_ ASCII Videos' (with links to 'Setting up Real-Time Network Monitoring', 'Measuring Public SMB Exposure', and 'Analyzing the Vulnerabilities for a Network'), and '</> Developer Access' (with links to 'How to Download Data with the API', 'Looking up IP Information', and 'Working with Shodan Data Files'). Below these cards is a 'QUICK LINKS' section with buttons for 'SETUP NETWORK MONITORING', 'BROWSE IMAGES', and 'MAP VIEW'. To the right of the quick links is a 'Filters Cheat Sheet' section, which includes a table with columns for 'Filter Name', 'Description', and 'Example'.

Shodan Maps Images Monitor Developer More...

SHODAN Explore Downloads Pricing Search... Account

Dashboard

Getting Started

- [What is Shodan?](#)
- [Search Query Fundamentals](#)
- [Working with Shodan Data Files](#)

LEARN MORE

>_ ASCII Videos

- [Setting up Real-Time Network Monitoring](#)
- [Measuring Public SMB Exposure](#)
- [Analyzing the Vulnerabilities for a Network](#)

VISIT THE CHANNEL

</> Developer Access

- [How to Download Data with the API](#)
- [Looking up IP Information](#)
- [Working with Shodan Data Files](#)

DEVELOPER PORTAL

// QUICK LINKS

SETUP NETWORK MONITORING

BROWSE IMAGES

MAP VIEW

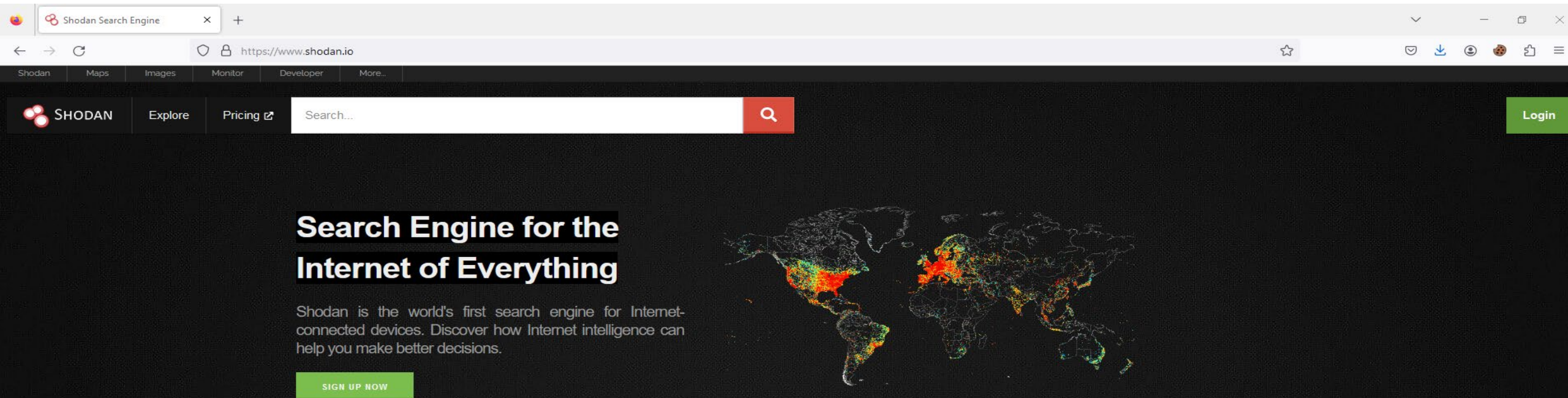
Filters Cheat Sheet

Shodan currently crawls nearly 1,500 ports across the Internet. Here are a few of the most commonly-used search filters to get started.

Filter Name	Description	Example
-------------	-------------	---------

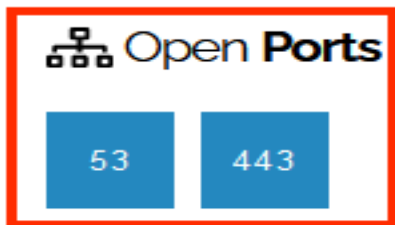
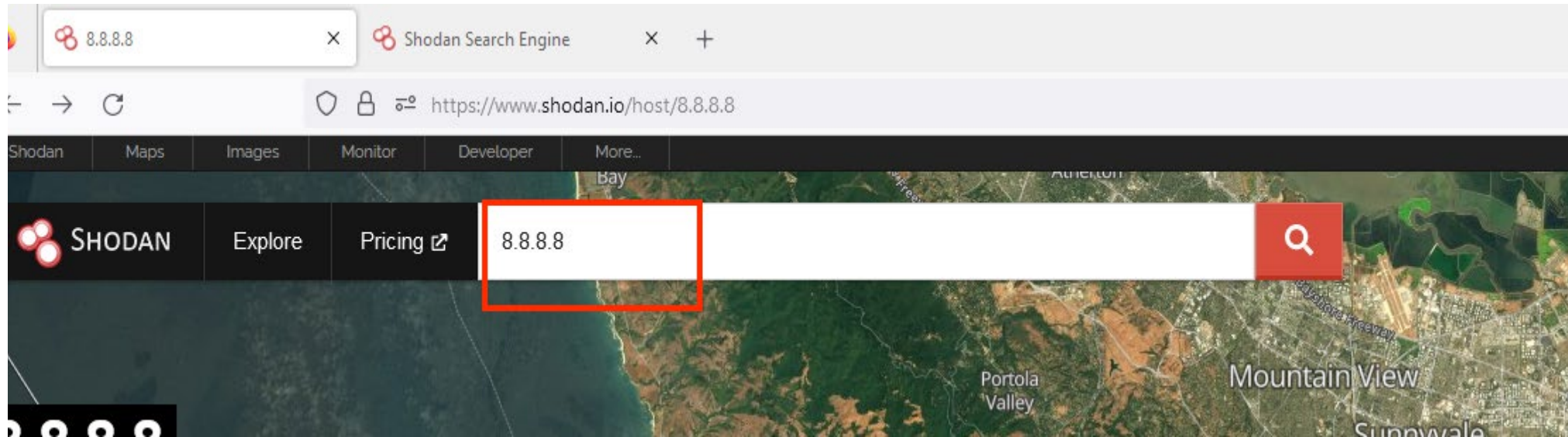
Introduction to Shodan

- Shodan is a popular search engine that allows users to discover and explore internet connected devices and services.
- Shodan works by scanning the internet for open ports and collecting data about the devices and services connected to those ports. This information, making it searchable for users.
- It is indexing information about devices such as webcams , routers, servers and other internet –connected devices.



Scanning using Shodan

When users search with some specific IP through Shodan , that scanners send out specific requests and probes to IP addresses attempting to establish connection with the target devices.



Banner Grabbing using Shodan

Banner Grabbing - - Once scanning successfully establish a connection with device or system, it collects the response which is called as “ banner”.

For example - - One website we searched through shodan which is hosted on the device that run a web service and Shodan would gather information by speaking with that web service

The screenshot displays the Shodan search results for the IP address 103.58.120.7. The interface includes a search bar at the top with the IP entered, and a map of Hyderabad in the background. The results are organized into several sections:

- General Information:** A table listing various details about the host.

General Information	
Hostnames	cdacb.in
Domains	CDACB.IN
Country	India
City	Hyderābād
Organization	National Informatics Centre Services Incorporated
ISP	National Informatics Centre Services Incorporated
ASN	AS141501
- Open Ports:** A section showing the open ports on the host.

Port
80
443
- Web Technologies:** A section listing the web technologies used by the host.

Technology
BOOTSTRAP
LIGHTBOX
- Apache httpd:** Two sections showing the banner response for the open ports 80 and 443.

```
HTTP/1.1 301 Moved Permanently
Date: Fri, 14 Jul 2023 21:28:37 GMT
Server: Apache
X-Frame-Options: SAMEORIGIN, SAMEORIGIN, SAMEORIGIN
Strict-Transport-Security: max-age=63072000; includeSubdomains; preload
Location: https://error/HTTP_BAD_REQUEST.html.var
Content-Length: 248
Connection: close
Content-Type: text/html; charset=iso-8859-1
```

Search Capabilities - - Users of Shodan can access the search functionality to query the indexed data based on specific criteria.

Example 1 - - We tried to find the webcams device which are available on the internet

Shodan

Maps

Images

Monitor

Developer

More...

SHODAN

Explore

Downloads

Pricing

webcams

Account

TOTAL RESULTS

158

TOP COUNTRIES



United States	102
Germany	19
France	5
United Kingdom	5
Austria	4
More...	

TOP PORTS

443	92
8081	24
9000	9
21	4
80	4
More...	

TOP ORGANIZATIONS

Mojohost	68
Mediacast	7
FRAZIER MOUNTAIN INTERNET	6

View Report

Browse Images

View on Map

Product Spotlight: Free, Fast IP Lookups for Open Ports and Vulnerabilities using [InternetDB](#)

35.171.151.160

ec2-35-171-151-160.compute-1.amazonaws.com

Amazon Technologies Inc.

United States, Ashburn

cloud eol-product

HTTP/1.1 200 OK

Server: nginx/1.20.0

Date: Tue, 18 Jul 2023 09:39:01 GMT

Content-Type: text/html

Content-Length: 863

Last-Modified: Sun, 26 Sep 2021 14:14:38 GMT

Connection: keep-alive

ETag: "6150804e-35f"

Accept-Ranges: bytes

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://w...

2023-07-18T09:39:01.758640

185.152.1.84

185-152-1-84.customers.pinemedia.net

Pine Media Ltd

United Kingdom, Sheffield

HTTP/1.0 200 OK

Content-type: text/html

Connection: close

Server: MJPG-Streamer/0.2

Cache-Control: no-store, no-cache, must-revalidate, pre-check=0, post-check=0, max-age=0

Pragma: no-cache

Expires: Mon, 3 Jan 2000 12:34:56 GMT

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://ww...

2023-07-18T08:17:31.865657



Example 2 - - Some of the search filters we can use

Shodan

Maps

Images

Monitor

Developer

More...

SHODAN

Explore

Downloads

Pricing ↗

port:22

TOTAL RESULTS

20,924,961

TOP COUNTRIES



United States	6,495,577
China	1,920,308
Germany	1,752,906
Brazil	1,033,677
Argentina	680,964

More...

TOP ORGANIZATIONS

Google LLC	1,651,041
DigitalOcean, LLC	1,223,246
Amazon Technologies Inc.	913,186
TELEFÓNICA BRASIL S.A	727,277
Telefonica de Argentina	633,451

More...

View Report

Browse Images

View on Map

Product Spotlight: Free, Fast IP Lookups for Open Ports and Vulnerabilities using [InternetDB](#)

2a01:4f8:c17:1b8::2
dedivirt43.your-server.de
[Hetzner Online GmbH](#)
Germany, Gunzenhausen

SSH-2.0-mod_sftp

Key type: ssh-rsa

Key: AAAAB3NzaC1yc2EAAAADAQABAAQDAdVVRhZHV0Y0UeEjQ4dGwKSh97No5FCVZU62tLBgTcRRE8eqfnekH2CEuYDxy1X1gv7QLq01SHUQ8cHcV9r8NSUQ341Eoo1UzUmpcIxx+PSaZpEKxc1DSOCHK6vTDF00E3Fv+ZaZpTuB8Nd9hTcssrKyasn/wuqxPvPdtNrwFXwhehvBEA68k1syXpXy17wdMyRF4n10956d6gofeqNSH5uRkzI4...

168.138.212.123
[Oracle Public Cloud](#)
Japan, Inzai

cloud

SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.2

Key type: ssh-rsa

Key: AAAAB3NzaC1yc2EAAAADAQABAAQgQD1ZldGZyRT81+O4CegZGq7uYCYewquveprDuPy1z6atyZucNNDxtJokeH5C5LRuS7gZ4GxKyroPjQN0Qdr7jUwTg10oFbwh8yxArTC8+GQgc3LLNBh6xfvXUGjeZU3SsbQ/+6KOZx5knQuU/1Hje2n9rhc4DrA2kZmfCY1H6YnSUXdvkxe7xgrT1ULsevc0vfnBLH9du...

174.138.112.56
[DigitalOcean, LLC](#)
Canada, Toronto

cloud

SSH-2.0-OpenSSH_7.2p2 Ubuntu-4ubuntu2.8

Key type: ssh-rsa

Key: AAAAB3NzaC1yc2EAAAADAQABAAQCG6AE2ydfI0+Q87j1V3yvtZehHDbYdN3def3o1bjNDBjNphNnXyMD3aaRayUnd6zTtjHpjNh57Q1ArFuW9Ae58xnqogBU1fz9OKGqAsGIzxFL/wisq3LPekvL1BD1hV372A+KHDI4Qez01SwnzSVEP0aezF4YrwGUY3R3ctcuuqjTq1How10a60+jqIwv+PSZFSC+...

104.197.92.176
176.92.197.104.bc.googleusercontent.com

SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.1

Example 3 - - you want find some network device which are available on internet like “netgear” or “switch” or “router”

ShodanMapsImagesMonitorDeveloperMore...

SHODANExploreDownloadsPricing

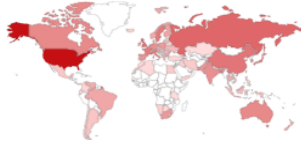
netgear

Account

TOTAL RESULTS

48,153

TOP COUNTRIES



United States	22,853
Russian Federation	2,622
United Kingdom	1,932
China	1,692
Sweden	1,411

More...

TOP PORTS

8443	19,060
8080	7,260
80	3,368
443	2,874
21	2,648

More...

View ReportView on Map

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401 Unauthorized

24.115.207.20
24.115.207.20.res-cmts.pbg.pld.net
[PenTeleData House Account](#)
 United States, Phillipsburg

self-signed

SSL Certificate

Issued By:
|- Common Name:
[www.routerlogin.net](#)
|- Organization:
NETGEAR

Issued To:
|- Common Name:
[www.routerlogin.net](#)
|- Organization:
NETGEAR

Supported SSL Versions:
TLSv1.2

HTTP/1.0 401 Unauthorized

WWW-Authenticate: Basic realm="NETGEAR R7850"
X-frame-options: SAMEORIGIN
Set-Cookie: XSRF_TOKEN=1222440606; Path=/
Content-type: text/html

2023-07-18T14:34:19.450088

HTTP 401 - Unauthorized

97.82.154.165
097-082-154-165.res.spectrum.com
[Charter Communications](#)
 United States, Morganton

HTTP/1.1 401 Unauthorized

Content-type: text/html
WWW-Authenticate: Basic realm="NETGEAR C7000v2"
Connection: close
Pragma: no-cache
Set-Cookie: XSRF_TOKEN=1390370; Path=/"

2023-07-18T14:29:13.388089

HTTP 401 - Unauthorized

76.178.129.75
cpe-76-178-129-75.natnow.res.r.com
[Charter Communications Inc](#)

HTTP/1.1 401 Unauthorized

Content-type: text/html
WWW-Authenticate: Basic realm="NETGEAR C7000v2"

2023-07-18T14:28:34.288467

Example 3 - - Search open port - 3389 in India

SHODAN

Explore

Downloads

Pricing

port:3389 country:"IN"

TOTAL RESULTS

57,746

TOP CITIES

Mumbai	33,449
Pune	6,888
Chennai	2,085
Delhi	1,942
Hyderābād	1,043

More...

TOP ORGANIZATIONS

Amazon Data Services India	23,676
Microsoft Corporation	8,563
Amazon.com, Inc.	2,855
Google LLC	1,404
Broadband Multiplay Project, O/o DGM BB, NOC B...	816

More...

TOP PRODUCTS

Remote Desktop Protocol	56,480
nginx	10
OpenSSH	9
Hikvision IP Camera	6
Microsoft ftpd	4

View Report

Browse Images

View on Map

Product Spotlight: Free, Fast IP Lookups for Open Ports and Vulnerabilities using [InternetDB](#)

122.165.4.14

abts-tn-static-014.4.165.122.airtelbr
oadband.in

ABTS Tamilnadu

India, Chennai

self-signed

SSL Certificate

Issued By:
|- Common Name:
SFTP-SERVER

Issued To:
|- Common Name:
SFTP-SERVER

Supported SSL Versions:
TLSv1, TLSv1.1, TLSv1.2

Diffie-Hellman Fingerprint:
RFC2409/Oakley Group 2

Remote Desktop Protocol
\x03\x00\x00\x13\xe\x00\x00\x124\x00\x02\t\x08\x00\x02\x00\x00

Remote Desktop Protocol NTLM Info:
OS: Windows 7/Windows Server 2008 R2
OS Build: 6.1.7601
Target Name: SFTP-SERVER
NetBIOS Domain Name: SFTP-SERVER
NetBIOS Computer Name: SFTP-SERVER
DNS D...

117.239.209.181

static.ill.117.239.209.181/24.bsnl.in

Ahmedabad Municipal Corporation,
Dangapith

India, Ahmedabad

SSL Certificate

Issued By:
|- Common Name:
EC2AMAZ-RPMQMJL

Issued To:
|- Common Name:
EC2AMAZ-RPMQMJL

Supported SSL Versions:
TLSv1, TLSv1.1, TLSv1.2

Remote Desktop Protocol
\x03\x00\x00\x13\xe\x00\x00\x124\x00\x02\xf\x08\x00\x02\x00\x00

Remote Desktop Protocol NTLM Info:
OS: Windows Server 2022
OS Build: 10.0.20348
Target Name: EC2AMAZ-RPMQMJL
NetBIOS Domain Name: EC2AMAZ-RPMQMJL
NetBIOS Computer Name: EC2AMAZ-RPMQMJL
DN...

35.154.199.1

ec2-35-154-199-1.ap-south-1.com
ute.amazonaws.com

Amazon Data Services India

India, Mumbai

cloud self-signed

SSL Certificate

Issued By:
|- Common Name:
EC2AMAZ-RPMQMJL

Issued To:
|- Common Name:
EC2AMAZ-RPMQMJL

Supported SSL Versions:
TLSv1, TLSv1.1, TLSv1.2

Remote Desktop Protocol
\x03\x00\x00\x13\xe\x00\x00\x124\x00\x02\x1f\x08\x00\x02\x00\x00

Remote Desktop Protocol NTLM Info:
OS: Windows Server 2022
OS Build: 10.0.20348
Target Name: EC2AMAZ-RPMQMJL
NetBIOS Domain Name: EC2AMAZ-RPMQMJL
NetBIOS Computer Name: EC2AMAZ-RPMQMJL
DN...

Gather Victim Identity Information – Using Shodan

Credentials (Use Shodan to find exposed login portals or services with authentication issues)

hostname:"cdac.in" http.title:"login"

hostname:"cdac.in" "Authorization required"

hostname:"cdac.in" "default password"

hostname:"cdac.in" "admin" port:22,21,8080

Email Addresses

hostname:"cdac.in" "email"

ssl.cert.subject.cn:"cdac.in"

http.html:"@cdac.in"

Employee Names

site:example.com filetype:pdf OR filetype:docx "employee list"

site:linkedin.com/in "company name" AND "employee"

intext:"meet the team" site:example.com

filetype:xls intext:"name" AND intext:"email"

Use site: to restrict to a specific domain.

Combine with filetype: to extract structured data (PDF, DOCX, XLS).

Use " " to force exact match phrases

Passive reconnaissance using WHOIS Lookups

- Publicly accessible database that contains registration information about domain names and IP addresses.
- It is maintained by domain registrars and regional internet registries (RIRs).
- Key Data Extracted from WHOIS Lookups
 - Domain Ownership Information
 - Domain Registration Details
 - Name Servers
 - IP Address and Hosting Provider
 - Geographical Information

Whois

Widely used tool in the footprinting

provides valuable domain registration information, including details about the domain owner, registrar, creation date, expiration date, and more.

Domain name registration and ownership are governed by the Internet Corporation for Assigned Names and Numbers (ICANN).

The typical information you can find in a Whois record includes -

1.Domain Information -

2.Nameservers -

3.Domain Status -

4.Contact Information

5.Domain Registrar

6.IP Address Block

7.Domain History

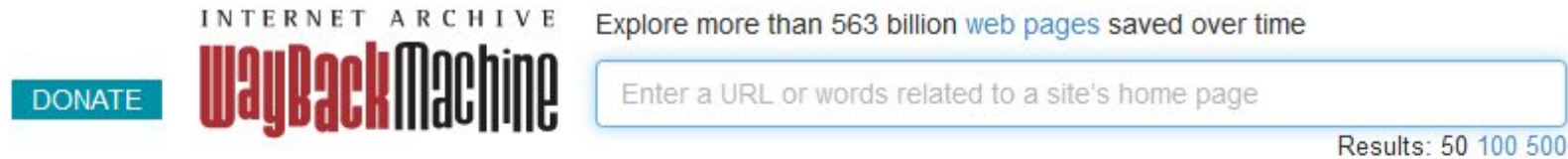
Whois References

- ARIN - [http - //ws.arin.net/whois](http://ws.arin.net/whois)
- RIPE NCC - [http - //www.ripe.net/whois/](http://www.ripe.net/whois/)
- APNIC - [http - //whois.apnic.net](http://whois.apnic.net)
- LACNIC - [http - //whois.lacnic.net](http://whois.lacnic.net)
- AfriNIC - [http - //whois.afrinic.net](http://whois.afrinic.net)

Passive reconnaissance using DNS Queries

- The Domain Name System (DNS) is the backbone of the internet, translating human-readable domain names (e.g., example.com) into IP addresses (e.g., 192.168.1.1).
- Passive DNS queries collect publicly available DNS records to map an organization's online infrastructure.

Information Gathering using “archive.org and netcraft.com”



What's that site running?

Using results from our **internet data mining**, find out the technologies and infrastructure used by any site.

<https://www.netcraft.com>

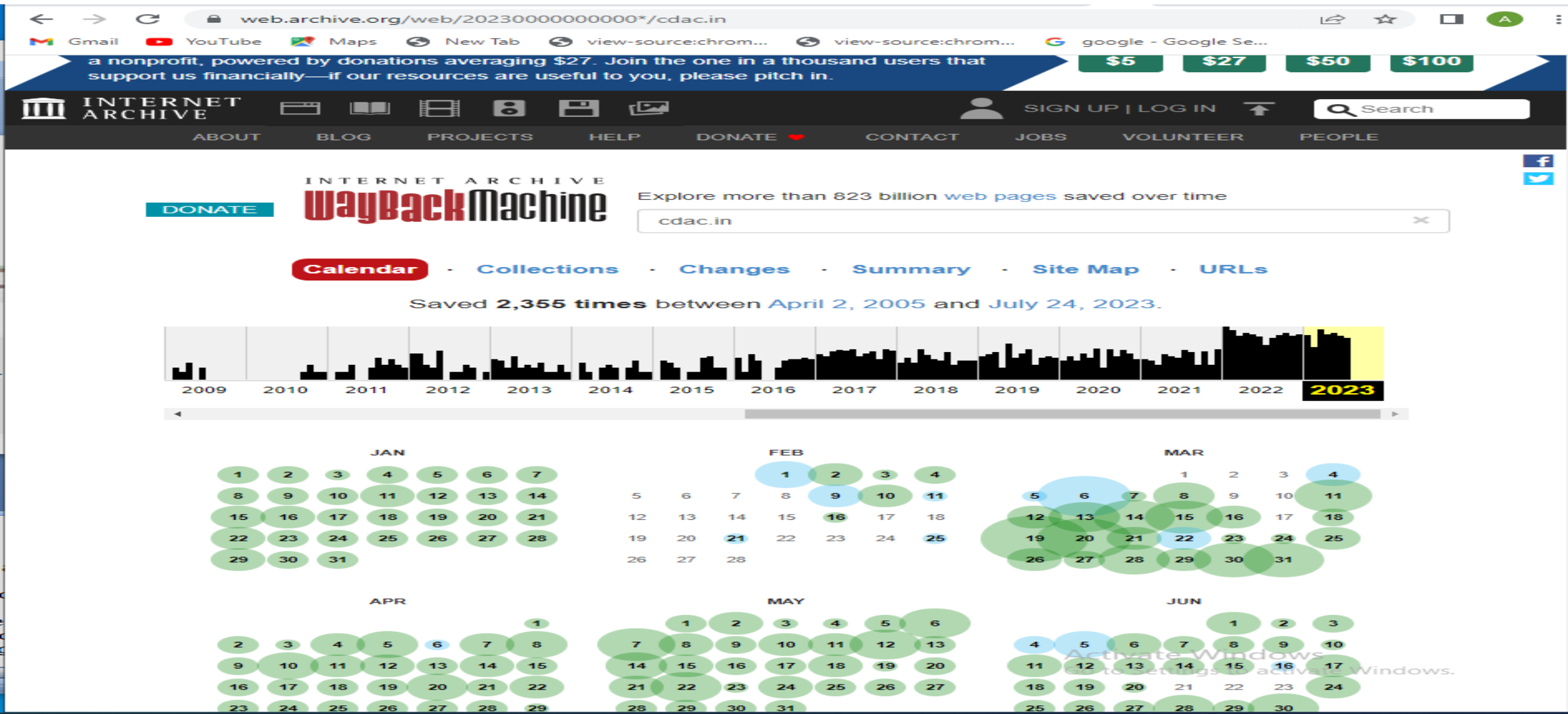


Audited by Netcraft

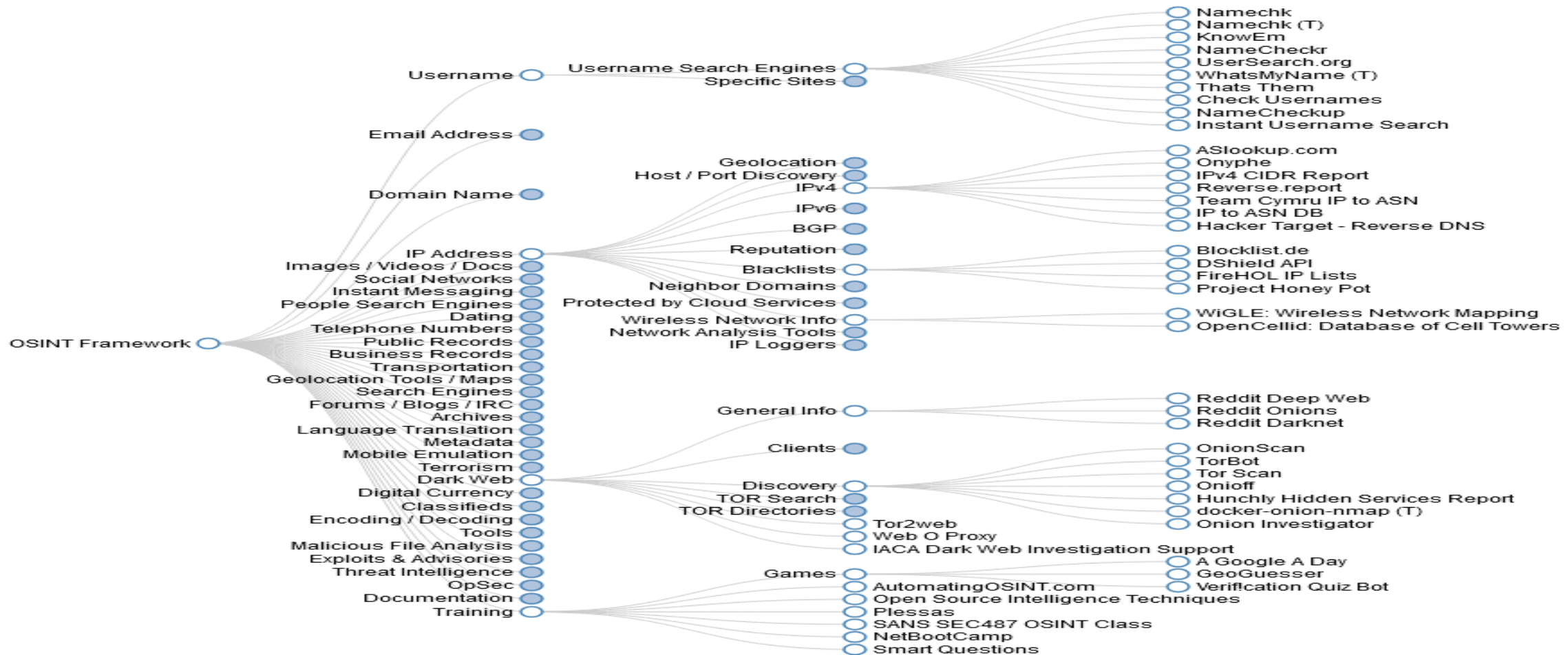
This site is Audited by Netcraft. **Get your site scanned for vulnerabilities**



Information Gathering using “archive.org”



OSINT Framework - <https://osintframework.com/>



Tools for Active Reconnaissance

- Dig
- Traceroute
- Pathping

Traceroute

It helps identify the network path and network hops between your system and a target system, revealing the intermediate routers and their IP addresses along the way.

- 1.Network Topology
- 2.Identify Network Devices
- 3.Troubleshooting
- 4.Latency Analysis
- 5.Identify Potential Entry Points
- 6.Network Security Assessment
- 7.Identify Redundant Paths

ping command

Is used to test the reachability of a host (computer or server) on an Internet Protocol (IP) network.

It also measures the round-trip time it takes for data packets to travel from your system to the target system and back.

- 1.Reachability
- 2.Network Latency
- 3.Identify IP Addresses
- 4.Identify Hostnames
- 5.Network Mapping
- 6.Time-to-Live (TTL) Analysis
- 7.Host Availability

Summary

- In this chapter you have learned the following -
- Phases of reconnaissance and Types of Footprinting
- How to Gather Information using google dork
- How to Gather Information using archive.org and netcraft.com
- How to Gather Information using Shodan and OSINT Framework
- The Tools for Footprinting